

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

2. CONTEXT DIAGRAM (LEVEL 0 DFD) The Context Diagram maps the linear client-server architecture and external API integration of the FitBuddy application.

- **Entity 1 (End-User):** Submits [Physical Metrics & Goals] via the UI.
- **Central Process (FitBuddy Backend):** Processes inputs, orchestrates the AI, and outputs a [Personalized Workout Plan].
- **External Entity (Google Gemini API):** Receives the [Formatted Prompt] and returns the [Structured Fitness Plan].

3. SYSTEM BREAKDOWN (LEVEL 1 DFD) This maps the sequential flow of data across the internal modules.

- **Process 1.0 (User Input):**
 - Captures POST request from the Jinja2 HTML form.
 - *Data Flow:* User UI → FastAPI (routes.py).
- **Process 2.0 (AI Orchestration):**
 - A crafted prompt containing the user's data is sent to the Gemini model (gemini-3.5-flash-lite).
 - *Data Flow:* FastAPI → Gemini API.
- **Process 3.0 (Data Persistence):**
 - SQLAlchemy maps the user details and the AI's response into relational tables (users and workout_plans).
 - *Data Flow:* FastAPI → SQLite (database.py).
- **Process 4.0 (UI Rendering):**
 - The backend injects the final customized plan into the HTML template for user viewing.
 - *Data Flow:* SQLite / FastAPI → Result UI.